

Data interpretation

COMPLETE LEARNING GUIDE

What You'll Learn

Data interpretation (DI) questions test whether a candidate can extract and compute accurately from tables, bar graphs, pie charts, line graphs, and short caselets under time pressure — the underlying arithmetic is usually simple, but the challenge is reading the right numbers off the right chart and knowing when to approximate versus calculate exactly. Most points are lost not from wrong formulas but from misreading axis scales, using the wrong base year for a percentage change, or spending too long on an exact calculation when a rounded estimate would eliminate three of four answer options instantly. This sub-module covers each major chart type with a fully worked numeric example, plus dedicated coverage of approximation techniques for speed and cross-category averaging, since DI sets are usually timed as a block and reward a consistent, fast reading-and-computing method over raw calculation speed alone.

1. Reading Tables Accurately

Table-based DI tests careful reading above all — identify the correct row/column intersection, and for 'percentage change' questions always divide by the EARLIER year's value (the base), not the later one.

- A company's sales (in ₹ lakh) were: 2019: 120, 2020: 150, 2021: 135, 2022: 180. Percentage change from 2021 to 2022 = $(180 - 135)/135 \times 100 = 33.33\%$.

2. Bar Graphs and Share of Total

Bar graphs commonly ask for one category's share of the total across all bars — sum every bar first, then divide the target bar by that sum.

- Five products A–E have sales of 40, 60, 50, 70, 30 (in units). Total = 250. Product D's share = $70/250 \times 100 = 28\%$.

3. Pie Charts: Degrees, Percentages, and Absolute Values

A pie chart's full circle is $360^\circ = 100\%$, so any sector's percentage = $(\text{sector degrees}/360) \times 100$, and converting to an absolute value requires multiplying that percentage by the given total.

- A 72° sector represents $72/360 \times 100 = 20\%$ of the total.

- If total students = 500 and the Engineering sector spans 90°, students in Engineering = $90/360 \times 500 = 125$.

4. Line Graphs and Trend Questions

Line graphs are typically used for average, trend, and rate-of-change questions across a sequence of points — read each point carefully off the y-axis before summing, since misreading even one point shifts the average.

- A line graph shows daily temperatures over 5 days: 20, 22, 19, 25, 23 (°C). Average = $(20+22+19+25+23)/5 = 109/5 = 21.8^\circ\text{C}$.

5. Caselet-Based DI

Caselets present data as a short paragraph rather than a chart, requiring you to first extract the relevant numbers into a mental (or scratch-paper) table before answering — treat this as a two-step process: extraction, then calculation.

- A shop sold Pen A: 200 units at ₹10, Pen B: 150 units at ₹15, Pen C: 100 units at ₹20. Total revenue = $2000+2250+2000 = ₹6250$. Pen B's revenue share = $2250/6250 \times 100 = 36\%$.

6. Approximation Techniques for Speed

When answer options are far apart, round the numbers to the nearest convenient value before dividing — this gets close enough to eliminate wrong options in a fraction of the time an exact calculation would take.

- Estimate $4875/62$ quickly: round to $4900/60 \approx 81.7$. The exact value (≈ 78.6) is close enough that if the options are 40, 79, 120, and 200, the approximation immediately identifies 79 without long division.

7. Comparing Averages and Ratios Across Categories

Questions that ask you to compare two data series (e.g., two companies' profits over several years) usually reduce to computing each series' average and then forming a ratio — do each average separately before comparing, rather than trying to compare running totals.

- Company X's profits over 3 years: 100, 120, 140 (avg = 120). Company Y's profits: 90, 110, 130 (avg = 110). Ratio of average profits X:Y = $120:110 = 12:11$.

Common Pitfalls

- Calculating a percentage change using the later year as the base instead of the earlier (original) year.
- Misreading a bar or line graph's y-axis scale, especially when it doesn't start at zero or uses non-uniform gridlines.

- Converting a pie chart sector to a percentage using the wrong total (forgetting the full circle is 360°, not 100).
- Spending time on exact long division in a DI set when the answer options are far enough apart for approximation to work.
- In caselets, answering from memory of the paragraph instead of re-extracting the exact numbers needed for that specific question.
- Comparing two data series by eyeballing totals instead of computing and comparing actual averages or ratios.

Key Takeaways

- Percentage change always divides by the earlier (base) value, never the later value.
- Pie chart sector % = $(\text{degrees}/360) \times 100$; multiply by the given total to get absolute values.
- A category's share of a bar/pie total = that category's value $\div$ sum of all categories.
- Approximate aggressively when answer options are far apart — exact precision wastes time DI sets don't allow.
- In caselets, extract numbers into a mini-table first, then compute — don't calculate from memory of the prose.
- Compute each series' own average before forming a comparison ratio between two data sets.

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